

$$\forall x [x \in \mathbb{R} \rightarrow \exists! y \in \mathbb{R} (x^2 y = x - y)]$$

$$x \in \mathbb{R}$$

$$\exists! y (y \in \mathbb{R} \wedge x^2 y = x - y)$$

$$\exists! y (x^2 y = x - y) \quad (\because \text{since it's implied that } y \text{ is real})$$

$$x^2 y = x - y$$

$$1) \exists y (x^2 y = x - y)$$

$$x^2 y + y = x$$

$$2) \forall s \forall t (x^2 s = x - s \wedge$$

$$x^2 t = x - t \rightarrow s = t)$$

$$y = \frac{x}{x^2 + 1} \quad (\because x^2 + 1 \neq 0)$$

$$\text{So } x^2 y = x^2 \left(\frac{x}{x^2 + 1} \right) = \frac{x^3}{x^2 + 1}$$

$$= \frac{(x^2 + 1)x - x}{x^2 + 1} = x - \frac{x}{x^2 + 1} \quad (\because \text{we figured this out by working with sides in Ruffini's rule})$$

$$= x - y$$

Let s & t be arbitrary $\in \mathbb{R}$

$$x^2 s = x - s \quad \text{--- (1)}$$

$$x^2 t = x - t \quad \text{--- (2)}$$

$$s = t$$

$$\text{From (1)} \rightarrow s = \frac{x}{x^2 + 1}$$

$$\& \text{ from (2)} \quad t = \frac{x}{x^2 + 1}$$

$$\text{So } s = t$$

2)

$$xy + x - y = 4y$$

$$x(y+1) = 4y + y$$

$$x = \frac{4y + y}{y+1}$$

$$x = \frac{4(y+1)}{y+1} = 4$$

$$\exists! x \forall y (xy + x - y = 4y)$$

$$1) \exists x \forall y (xy + x - y = 4y)$$

$$2) \forall s \forall t [y(xy + s - y = 4y)$$

$$\wedge \forall y (xy + t - y = 4y) \rightarrow s = t]$$

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Let s & t be arbitrary

$$\text{Suppose } s/n = s - n \quad \text{--- (1)}$$

$$s = t$$

$$t/n = t - n \quad \text{--- (2)}$$

$$\textcircled{1} \rightarrow s = \frac{n^2}{n-1}$$

$$\textcircled{2} \rightarrow t = \frac{n^2}{n-1} \quad \text{so } s = t$$

$$4) \forall x \in \mathbb{R} [x \neq 0 \rightarrow \exists! y \forall z (zy = z/x)]$$

$$n \in \mathbb{R}$$

$$n \neq 0$$

$$\exists! y \forall z (zy = z/n)$$

$$1) \exists y \forall z (zy = z/n)$$

$$zy = zx$$

$$\text{so } y = \frac{z}{x} \times \frac{1}{z}$$

$$\text{so } y = \frac{1}{x} \quad (\text{as } x \neq 0, \text{ so } \frac{1}{x} \text{ is defined})$$

$$\text{so } zy = z \frac{1}{x} = z/x$$

$$2) \forall s [\forall z (zs = z/x) \rightarrow y = s]$$

Suppose $\forall z (zs = z/x)$ & s is arbitrary that satisfies this eq.

$\therefore$ If we have proved $y = \frac{1}{x}$ for arbitrary z then we can use any value of z to show again the same result. Then in particular taking $z = 1$, we get $s = \frac{1}{x}$.

$$\text{so } s = y$$

In first part, we ~~had~~ defined $y = \frac{1}{x}$ & $x \neq 0$. So there was no undefined division. If in the second part, we did same then $s = \frac{z}{x}$, were there is division by x , so x must not be 0. So we have to make sure that's why we took $x \neq 0$.

For y , we found a value for arbitrary z , for s , we assumed that the eq. holds for all z , so we are given that eq. holds for every z & we can therefore use any value of z .

5) (a) $U!F = \{x \mid \exists! A (A \in F \wedge x \in A)\}$

Prove that $U!F \subseteq UF$

$$\forall x [x \in U!F \rightarrow x \in UF]$$

Supp $x \in U!F = \exists! A (A \in F \wedge x \in A)$

$x \in UF$

$\exists A (A \in F \wedge x \in A) \quad \text{--- ①}$

$\exists A (A \in F \wedge x \in A)$

$\forall B \forall C [(B \in F \wedge x \in B) \wedge (C \in F \wedge x \in C) \rightarrow B = C] \quad \text{--- ②}$

Goal follows from 1st given

(b) $U!F = UF$ iff F is pairwise disjoint

$\rightarrow$ Suppose $U!F = UF$

$\forall A \in F \forall B \in F (A \neq B \rightarrow A \cap B = \emptyset)$

$\forall x [x \in U!F \rightarrow x \in UF] \quad \text{--- ①}$

$\forall x [x \in UF \rightarrow x \in U!F] \quad \text{--- ②}$

① & ② means $\forall x [\exists! A (A \in F \wedge x \in A) \leftrightarrow \exists A (A \in F \wedge x \in A)]$

which means

$\forall x [\exists A (A \in F \wedge x \in A) \wedge (\forall C \forall D [(C \in F \wedge x \in C) \wedge (D \in F \wedge x \in D)] \rightarrow C = D) \leftrightarrow \exists A (A \in F \wedge x \in A)]$

Let $A \in F$ & $B \in F$ be arbitrary &

$\exists A (A \in F \wedge x \in A)$

suppose $A \neq B$.

③

we will prove by contradiction

Goal: Contradict

suppose $A \cap B \neq \emptyset$

then $\exists y [y \in A \wedge y \in B] \quad \text{--- ④}$

so $y \in A \wedge y \in B$

Since $A \in F$ & $y \in A$ so from ③ $\leftarrow$ direction we can conclude

$\exists A (A \in F \wedge y \in A) \wedge \forall C \forall D [(C \in F \wedge y \in C) \wedge (D \in F \wedge y \in D)] \rightarrow C = D$

Since $A \in F \wedge y \in A$ & since $B \in F \wedge y \in B$ so $A = B$

But this is contradiction on over $A \neq B$.

$$\leftarrow \forall A \in F \forall B \in F (A \neq B \rightarrow A \cap B = \emptyset) \quad (b) \quad U \setminus F = U \setminus F$$

We already know from theorem in part (a) that $U \setminus F \subseteq U \setminus F$ for every F . We only need to show that $U \setminus F \subseteq U \setminus F$.

$$\forall x [x \in U \setminus F \rightarrow x \in U \setminus F]$$

$$x \in U \setminus F$$

$$x \in U \setminus F$$

$$\exists! A (A \in F \wedge x \in A)$$

$$\exists A (A \in F \wedge x \in A) \quad (a)$$

$$1) \exists A (A \in F \wedge x \in A)$$

local ① follows from (a)

$$2) \forall C \in F (C \in F \wedge x \in C) \rightarrow A = C$$

$$\text{For } ② \text{ suppose } C \in F \wedge x \in C \quad A = C$$

since $x \in A$ by (a) and $x \in C$ for arbitrary x , so

$$A \cap C \neq \emptyset \text{ so from (b) by modus tollens } A = C$$

$$6) (a) \exists! A \in P(U) \forall B \in P(U) (A \cup B = B)$$

$$1) \exists A \in P(U) \forall B \in P(U) (A \cup B = B)$$

$$2) \forall C [C \in P(U) \wedge \forall B \in P(U) (C \cup B = B) \rightarrow C = \emptyset]$$

1) Let $A = \emptyset$ then $\emptyset \in P(U)$ since $\emptyset \subseteq A$ for every set A .

And clearly for every $B \in P(U)$ $\emptyset \cup B = B$.

2) Suppose $C \in P(U) \wedge \forall B \in P(U) (C \cup B = B)$ then in particular take $B = \emptyset$ we have $C \cup \emptyset = \emptyset$. But also since $C \cup \emptyset = C$, so $C = \emptyset$ so $C = A$.

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- (b) $\exists! A \in P(U) \forall B \in P(U) (A \cap B = A)$
1) $\exists A \in P(U) \forall B \in P(U) (A \cap B = A)$
2) $\forall C (C \in P(U) \wedge \forall B \in P(U) (C \cap B = C) \rightarrow C = A)$

Let $A = U$. $U \in P(U)$. To see that $\forall B \in P(U)$

- 1) we will use an already proved theorem that if $A \subseteq B$ then $A \cup B = B$.

Let B be arbitrary & suppose $B \in P(U)$ then $B \subseteq U$ since $B \subseteq U$ so $B \cup U = U$.

- 2) Suppose $C \in P(U) \wedge \forall B \in P(U) (C \cap B = C)$
then in particular taking $B = U$, we get $C \cap U = C$
but since $C \cup U = U$ as $C \subseteq U$ ($\because C \in P(U)$)
so $C = U$

- 7) $\exists! A \in P(U) \forall B \in P(U) (A \cap B = B)$

- 1) Let $A = U$. $U \in P(U)$. And $\forall B \in P(U) (U \cap B = B)$
This is previously proved theorem if $B \subseteq U$ then $B \cap U = B$

- 2) Suppose $C \in P(U) \wedge \forall B \in P(U) (C \cap B = B)$
then taking $B = U$, we get $C \cap U = U$. But since $C \subseteq U$ so $C \cap U = C$. so $C = U$.

- (b) $\exists! A \in P(U) \forall B \in P(U) (A \cap B = A)$

- 1) Let $A = \emptyset$. $\emptyset \in P(U)$ & $\forall B \in P(U) (\emptyset \cap B = \emptyset)$

- 2) Suppose $C \in P(U) \wedge \forall B \in P(U) (C \cap B = C)$

then in particular if $B = \emptyset$, then $C \cap \emptyset = C$. But since $C \cap \emptyset = \emptyset$, so $C = \emptyset$.

$$8) \forall A \in P(U) \exists! B \in P(U) \forall C \in P(U) (C \cap A = C \cap B)$$

let A be arbitrary & sup $A \in P(U)$

$$\exists! B \in P(U) \forall C \in P(U) (C \cap A = C \cap B)$$

$$\exists! B \in P(U) \forall C \in P(U) (\forall x (x \in C \cap x \notin A \rightarrow x \in C \cap x \in B))$$

let $B = C \cap A$ since $A \in P(U)$ & $C \in P(U)$ so $B \in P(U)$
 already proved theorem ($\therefore$ if $A \in P(U) \wedge B \in P(U)$ then $A \cap B \in P(U)$)

To see that $C \cap A = C \cap B = C \cap (C \cap A)$

let $x \in C \cap A$

$$\text{iff } x \in C \wedge x \in A$$

$$\text{iff } x \in C \wedge x \in (C \cap A) \quad (\therefore \text{ idempotent law})$$

$$\text{iff } x \in C \wedge (x \in C \wedge x \in A)$$

$$\text{iff } x \in C \wedge x \in C \wedge x \in A$$

$$\text{iff } x \in C \cap C \wedge x \in A$$

$$\text{iff } x \in C \cap B$$

To see that x is unique. Suppose $B \in P(U) \forall C \in P(U)$
 $(C \cap A = C \cap B)$

We cannot take $B = C \cap A$ because for a particular B , we have to make the statement true for every C . So B should not depend on C . If it does, it will contradict the $\forall C$ quantification. (We want declare a unique $B = C \cap A$ & then take C arbitrarily. But if C changes, B also have to change & in no more unique)

As $C \cap A = C \cap B$ then $x \in C \wedge x \notin A = x \in C \wedge x \in U \wedge x \notin A$
 $\therefore$ see that $C \subseteq U$ as $C \in P(U)$ so every $x \in C \cap (U \setminus A)$
 $x \in C$ also $x \in U$. So $U \setminus A$ works.

let $B = U \setminus A$ since $A \in P(U)$ & $U \in P(U)$ so $U \setminus A \in P(U)$

To see that $C \cap A = C \cap B = C \cap (U \setminus A)$

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$$\begin{array}{ll} \rightarrow \text{Suppose } x \in C \setminus A & x \in C \cap (U \setminus A) \\ x \in C \cap x \notin A & x \in C \cap x \in U \cap x \notin A \end{array}$$

since $C \subseteq U$ so $x \in U$

$$\leftarrow \text{Suppose } x \in C \cap (U \setminus A) \quad x \notin C \setminus A$$

so $x \in C \cap x \notin A$ so $x \in C \setminus A$

2) To see that B is unique. Suppose $D \in P(U) \forall C \in P(U)$
 $C \setminus A = C \cap D$. Then in particular taking $C = U$, we
 get $U \setminus A = U \cap D$ but since $D \subseteq U$, so $D \cap U = D$
 so $U \setminus A = D = B$.

$$(b) A \in P(U) \exists! B \in P(U) \forall C \in P(U) (C \cap A = C \cap B)$$

Let $A \in P(U)$

$$C \cap A = C \setminus B$$

$$\exists B \in P(U) \forall C \in P(U) (C \cap A = C \cap B)$$

$$\forall D \in P(U) \forall C \in P(U) (C \cap A = C \cap D)$$

$$\rightarrow D = B$$

Let $B = U \setminus A$

Clearly $B \in P(U)$

Now to show $C \cap A = C \cap B = C \setminus (U \setminus A)$

$$\rightarrow \text{Supp } x \in C \cap A$$

$$x \in C \cap B$$

$$x \in C \cap x \in A$$

$$x \in C \setminus (U \setminus A)$$

Clearly goal follows

$$x \in C \cap (x \notin U \vee x \in A)$$

$$\leftarrow \text{Supp } x \in C \setminus (U \setminus A)$$

$$x \in C \cap (x \notin U \vee x \in A)$$

$$x \in C \cap A$$

We can rule out possibility that $x \notin U$

$$x \in C \cap x \in A$$

because we are only considering elements

that $\in$ set U . so $x \in C \cap x \in A$.

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So we can
write $\{x\}$ possibility of $x \in U$.

In this question, we cannot take $P(A) = A$ because we cannot rule out possibility that $x \notin A$. But we can write $U \setminus A$. Since all sets under discussion are subsets of U , so $U \setminus A$ relation universal set must belong to U .

2) To show that B is unique $\supset D$ be arbitrary $\in P(U)$
 $\forall C \in P(U) (C \cap A = C \cap B)$.

In particular taking $C = U$ $D = U \setminus A$

we get $U \cap A = U \cap D$

Since $A \subseteq U$ so $U \cap A = A$

so $A = U \cap D$ ①

$\supset x \in D \cdot A \subseteq U$ so $x \in U$.

$x \in U \setminus A$

① $\rightarrow x \in A \rightarrow x \in U \cap D$

$x \in U \setminus A \rightarrow x \notin A$

$x \in A \rightarrow x \in U \cap D$ ②

but since $x \in D$, so by ② we have $x \in A$.

Now suppose $x \in U \setminus A$

$x \in D$

$x \in U \setminus A$

~~Suppose $x \notin D$. Then since $D \subseteq U$~~

we have $A = U \cap D$ so $x \notin A$

so $x \notin U \vee x \in D$ but since $x \in U$ so $x \in D$

9) (a) Prove that symmetric difference is commutative
 $A \oplus B = B \oplus A$.

(a)

$\exists! x \forall y (A \oplus x = y)$

$\exists! x \forall y (A \oplus x = y)$

$\forall y \exists! x (A \oplus x = y) \rightarrow y = x$

Let $X = \emptyset$ we have to show

Let A be arbitrary

$\forall A (A \oplus \emptyset = A)$

$\supset x \in A \oplus \emptyset$

$x \in A$

(Clearly) $A \oplus \emptyset = (A \cap \emptyset) \cup (\emptyset \cap A) = A \cup \emptyset = A$

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We can prove uniqueness by $\forall Y \forall Z (P(Y) \wedge P(Z) \rightarrow Y=Z)$ using commutative law for symmetric difference. But let us for now stick to our approach.

supp ~~$X \oplus Y = A$~~

As ~~$A \oplus Y = A$~~ then in particular taking $A = \emptyset$, we get $\emptyset \oplus Y = \emptyset$

$A \oplus Y = (\emptyset \oplus Y) \cup (Y \setminus \emptyset) = Y$ But since ~~$A \oplus Y = A$~~ so $Y = A$.
 $\emptyset \oplus Y = Y$ & $\emptyset \oplus Y = \emptyset$ so $Y = \emptyset = A$

(b) $\forall A \exists! B (A \oplus B = X)$ or
 $\forall A \exists! B (A \oplus B = \emptyset)$ As $\emptyset$ is identity etc

1) let A be arbitrary

let B be A . Clearly $(A \setminus A) \cup (A \setminus A) = \emptyset \cup \emptyset = \emptyset = X$

2) Supp there are C & D such that $\forall A$

$A \oplus C = \emptyset$ & $A \oplus D = \emptyset$

letting $A = \emptyset$ in ① we get $\emptyset \oplus C = \emptyset$ ($\because$ as A is arbitrary)

letting $A = \emptyset$ in ②, we get $\emptyset \oplus D = \emptyset$ But since $\emptyset \oplus D = D \oplus \emptyset$

so $C = D$ But since $A \oplus C = \emptyset$, so $C = \emptyset$. Similarly

$D = \emptyset$, so $C = D$

(c) $\forall A \forall B \exists! C (A \oplus C = B)$

let A & B be arbitrary

let $C = A \cup B$

Wrong (we have to let A be arbitrary)

Suppose there are C & D such that $\forall A$

$A \oplus C = \emptyset$ & $A \oplus D = \emptyset$

We also know $A \oplus \emptyset = A$ for any A by previous theorem

Now $\emptyset \oplus \emptyset = \emptyset \rightarrow$ putting $\emptyset = A \oplus D$ we get $\emptyset \oplus (A \oplus D) = (\emptyset \oplus A) \oplus D$

$= (A \oplus \emptyset) \oplus D = A \oplus D = \emptyset$ so $C = D$.

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$$(c) \quad \forall A \forall B \exists! C (A^3 C = B)$$

Let $C: A^3 B$. Then $A^3 C = A^3 (A^3 B) = (A^3 A)^3 B = \phi^3 B = B$.

To see uniqueness, suppose there are D & E that satisfy this property. i.e. $A^3 D = B$ & $A^3 E = B$.

Note that $A^3 D = A^3 E = B$.

$$\begin{aligned} \text{Now } D &= D^3 \phi = D = D^3 (A^3 A) = (D^3 A)^3 A \\ &= (A^3 D)^3 A = (A^3 E)^3 A = (E^3 A)^3 A = E^3 (A^3 A) \\ &= E^3 \phi = E \quad \text{so } D = E. \end{aligned}$$

We could have used this theorem: if $A^3 B = A^3 C$ then $B = C$.

Proof.

$$\forall x [(x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin A)] \rightarrow [(x \in A \wedge x \notin C) \vee (x \in C \wedge x \notin A)] \quad (1)$$

We have to prove that $B = C$.

We will prove by contraposition. Suppose $B \neq C$.

$$\exists x [(x \in B \wedge x \notin C) \vee (x \in C \wedge x \notin B)]$$

$$(x \in B \wedge x \notin C) \vee (x \in C \wedge x \notin B)$$

Suppose $(1) \quad x \in B \wedge x \notin C$. Suppose case 1) $x \in A$. Then by side of (1) $(x \in A \wedge x \notin B) \vee x \in B \wedge x \notin A$ $x \in A \wedge C$ so $x \in A^3 C$.

By $\leftarrow$ side of (1). We can conclude that $(x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin A)$.

Clearly considering any of these 2 lead to contradiction.

Case 2) $x \notin A$. Then $x \in B \wedge A$ so $x \in A^3 B$. The $\rightarrow$ side of (1)

we can conclude that $(x \in A \wedge x \notin C) \vee (x \in C \wedge x \notin A)$ clearly both of these are false, so $x \notin A^3 C$.

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Now supp $x \in C \cap x \notin B$. Case 1) supp $x \in A$. So $x \in A \cap B$ so $x \in A \cap B$. So by $\rightarrow$ side of ①. We conclude the $\rightarrow$ side.
But clearly both of the are false.

Case 2) $x \notin A$. then $x \in C \cap A$ so $x \in A \cap C$. So by $\leftarrow$ side of ① we conclude it, but both of the are false.

So $A \cap B \neq A \cap C$ when we supposed that $B \neq C$. This completes proof by contraposition.

$$(d) \quad \forall A \exists! B \subseteq A \forall C \subseteq A (B \cap C = A \cap C)$$

Let A be arbitrary set

Let $B = A$. Clearly $A \subseteq A$. Suppose $C \subseteq A$. Now $B \cap C$

$$= A \cap C = (A \cap C) \cup (C \cap A). \text{ As } C \cap A = \emptyset \text{ because } C \subseteq A \text{ so}$$

$$A \cap C \cup \emptyset = A \cap C.$$

To show that $B = A$ is unique. Suppose there are sets $D \neq E$ that satisfy this property. i.e. $D \cap C = A \cap C$ & $E \cap C = A \cap C$.

Since $D \cap C = E \cap C$ so $D = C$. ($\because$ By theorem we proved before)

We can also show uniqueness by taking $C = \emptyset$ in particular

$$\text{then } D \cap \emptyset = A \cap \emptyset \rightarrow D = A$$

$$10) \quad \forall F (\cup F = A \rightarrow A \in F)$$

A has exactly 1 el.

$$\forall F (\forall x (x \in \cup F \leftrightarrow x \in A) \rightarrow A \in F)$$

$$\exists! y (y \in A)$$

$$\forall F (\exists x \in F (x \in A) \rightarrow A \in F)$$

It's hard to get anywhere if we use arbitrary elements on this proof. We are not given if A is non empty

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So we first have to show that $\emptyset \neq A$.

To see this suppose $A = \emptyset$. Suppose $F = \emptyset$, then $UF = \emptyset$ so $UF = A$ but $A \neq F$ as F is empty. This is a contradiction so A must have at least one element.

$\exists x (x \in A)$

Suppose $x \in A$ is an element of A . $\forall y [y \in A \rightarrow y = x]$
To see that x is unique & only element of A .

Suppose $y \in A$

$y = x$

We will prove by contradiction

Suppose $y \neq x$

Contradiction

So $A = \{x, y\}$ $x \neq y$

• $\forall F \rightarrow \emptyset$ To contradict the universal quantifier, we only need one counterexample.

Suppose $F = \{\{x\}, \{y\}\}$ then $UF = \{x, y\} = A$

but $A = \{x, y\} \notin F$. So this is a contradiction over $x \neq y$, so x must be equal to y .

$\exists F$

1) $\forall G \subseteq F (UG \subseteq F)$

$\exists! A (A \in F \wedge \forall B \in F (B \subseteq A))$

$\neg (G \subseteq F \rightarrow UG \subseteq F) \rightarrow \emptyset$

$\exists A (A \in F \wedge \forall B (B \in F \rightarrow B \subseteq A))$

Let $A = UF$ ($\therefore$ clearly $UUF = U \emptyset = \emptyset$)
 $\neg (UF \subseteq F \rightarrow UUF \subseteq F) \rightarrow \emptyset$
 $\neg (UF \subseteq F \rightarrow UUF \subseteq F) \rightarrow \emptyset$

$UF \subseteq F$ - from 1)

$C = D$

Suppose $B \in F$

$B \subseteq A$

For any $B \in F \rightarrow B \subseteq UF$ ($\therefore$ already proved known)

So $B \subseteq UF = A$

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To use uniqueness, suppose there are C & D that satisfy property $\forall B (B \in E \rightarrow B \subseteq C) \wedge \forall B (B \in E \rightarrow B \subseteq D)$
 Taking $B = C$ in first we get $C \subseteq C$ & taking $B = C$ in second we get $C \subseteq D$ so $C = D$

12)(a) $\exists x_1 \exists x_2 [P(x_1) \wedge P(x_2) \wedge x_1 \neq x_2] \wedge \forall y [P(y) \rightarrow y = x_1 \vee y = x_2]$

This models all cases including empty domain, exactly 1, exactly 2 & more than 2.

(b) Find x_1 & x_2 & show that $x_1 \neq x_2$

Then suppose $P(y)$ & prove that $y = x_1 \vee y = x_2$

(c) $x^3 = x^2$ has exactly 2 solutions

> Actually a better & more clear is

$$\exists x_1 (P(x_1) \wedge \exists x_2 (P(x_2) \wedge x_1 \neq x_2 \wedge \forall y \dots))$$

Anyway let $x_1 = 1$ & $x_2 = 0$

Clearly $1^3 = 1^2$ & $0^3 = 0^2$

also $1 \neq 0$ so $x_1 \neq x_2$

Suppose $y^3 = y^2$ for arbitrary y

$$y = x_1 \vee y = x_2$$

then $y^3 = y$ & $y^2 - y = 0$

$$\text{so } y(y-1) = 0 \text{ so } y = 0$$

$$\text{or } y-1 = 0 \rightarrow y = 1$$

— if we consider division by 0 then if we want to divide by $y-1$ then $y \neq 1$ which leads to $y = 0$
 $y^3 = y^2$ so $y^3 - y^2 = 0$
 $y^2(y-1) = 0$
 leads to either

13)(a)

$$\exists c \exists! x (x^2 + 3x + c = 0)$$

Note that this is $\exists c [\exists! x (x^2 + 3x + c = 0) \wedge \forall d (P(x=d) \rightarrow d=c)]$

So we will have to find c first then prove that x is unique against it & then prove that c is unique.

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let $c = \frac{9}{4}$ ($\dots$ found by using quadratic formula $-3 \pm \sqrt{3^2 - 4c}$
 if $3^2 - 4c = 0$ then $c = \frac{9}{4}$) 2

then $x^2 + 3x + \frac{9}{4} = 0$ has only one solution

which is $x = -\frac{3}{2}$ $(-\frac{3}{2})^2 + 3(-\frac{3}{2}) + \frac{9}{4} \rightarrow \frac{9}{4} - \frac{9}{2} + \frac{9}{4} = \frac{9}{4} - \frac{18}{4} + \frac{9}{4} = 0$ = P.H.

To see that this solution is unique. Suppose some x_1 satisfies the eq. then $x_1^2 + 3x_1 + \frac{9}{4} = 0$. Applying quadratic formula to this eq leads to only one solution that is $-\frac{3}{2}$ so $x_1 = x$

To see now that c is unique suppose there is D such that $(-\frac{3}{2})^2 + 3(-\frac{3}{2}) + D = 0$ $\frac{9}{4} - \frac{9}{2} + D = 0$

so $D = \frac{9}{4} = c$. (if c is not $\frac{9}{4}$ then x must satisfy that)

(b) $\sim \exists! x \exists! c (x^2 + 3x + c = 0)$

let x be arbitrary suppose $x^2 + 3x + c = 0$

then $x(x+3) = -c$ or $c = -x(x+3)$ so c is not unique & for every x , c is different as c depends on x .